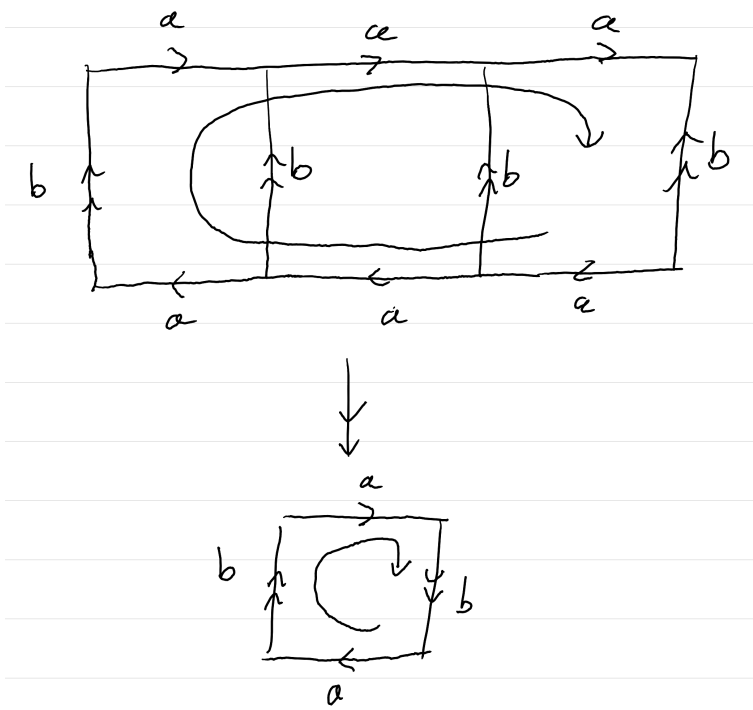
Revised Problems

Problem (Homework 17, Problem 1)**:** Give two examples of a nontrivial covering space of the Klein bottle, one normal and one non-normal.

Solution: As discussed in the original submission, we observe that the torus can be viewed as a two-sheeted cover of the Klein bottle, and since the number of sheets is equal to the index of $p_*\left(\pi_1\left(\widetilde{X}, \widetilde{x}_0\right)\right)$ in $\pi_1(X, x_0)$, we have that this is a normal cover.

To find a non-normal cover, we consider the three-sheeted covering space given below. This covering space lifts the loop given by a to three copies of itself, while it lifts b to one copy of itself. In particular, its fundamental group corresponds to the subgroup of the Klein bottle group $\left\langle a, b | abab^{-1}\right\rangle$ given by $H = \left\langle a^3, b\right\rangle$. Yet, this subgroup is not normal since $b \in H$, but $aba^{-1} = a(aba)a^{-1} = a^2b \notin H$.



Problem (Homework 17, Problem 2)**:** Let $p\colon \widetilde{X} \to X$ be a covering space with X path-connected, locally path-connected, and semilocally simply connected, but $\widetilde{X}$ is not necessarily path-connected. Prove that the path-components of $\widetilde{X}$ are in bijection with the orbits of the action of $\pi_1(X, x_0)$ on the fiber $p^{-1}(x_0)$.

Solution: We have that $\pi_1(X, x_0)$ acts on the fiber $p^{-1}(x_0)$ via lifting. That is, if γ is a loop based at x_0 in X, then $[\gamma]$ lifts to some path $\widetilde{\gamma}$ taking $\widetilde{x_0} \in p^{-1}(x_0)$ to some point $\widetilde{x_1} \in p^{-1}(x_0)$, and the action takes $\widetilde{x_0} \mapsto \widetilde{x_1}$.

Suppose Y is a path-component of $\widetilde{X}$. We know that $\pi_1(X, x_0)$ acts transitively on the restriction of the fiber $p^{-1}(x_0)$ when restricted to Y, since a path-component of a covering space is a path-connected covering space. In particular, if $\widetilde{x_0} \in p^{-1}(x_0) \cap Y$ is any element, then the orbit of $\widetilde{x_0}$ consists of all elements of $p^{-1}(x_0)$ that are contained in Y. Since the orbit of any element of $p^{-1}(x_0)$ is the path-component that element lies in, it follows that the orbits of the action are in one-to-one correspondence with the path-components of $\widetilde{X}$.

Current Problems

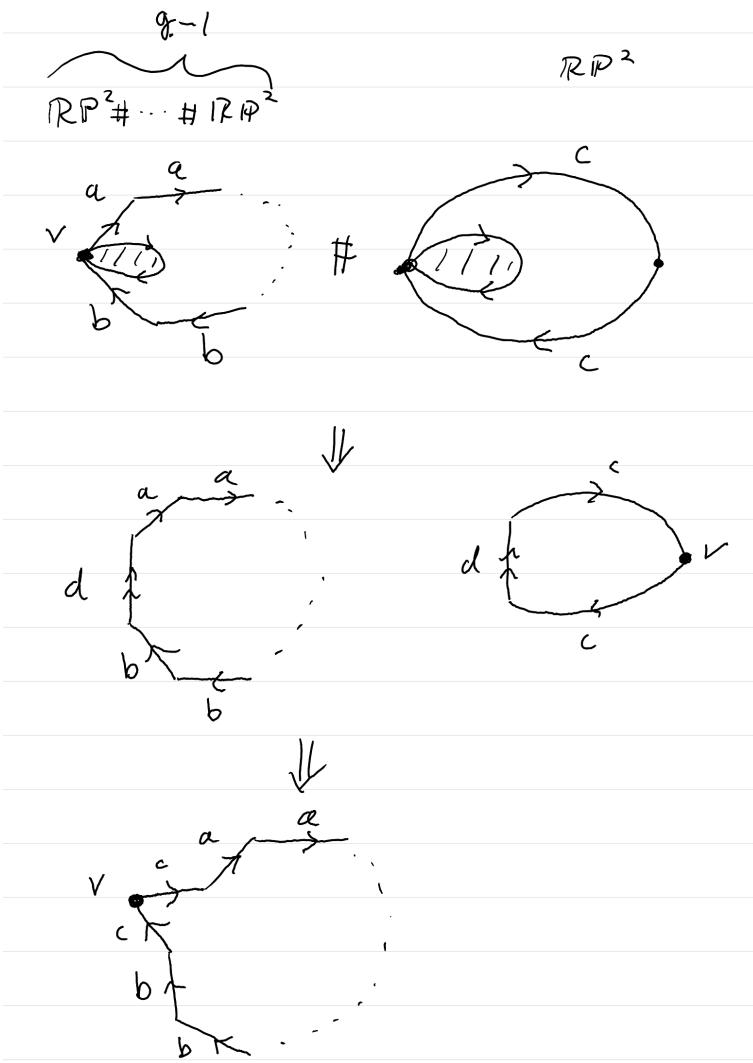
Problem (Problem 1)**:** Compute the homology groups of

$$\underbrace{\mathbb{RP}^2 \# \cdots \# \mathbb{RP}^2}_{g}$$

for any $g \geq 1$.

Solution: We will label the surface as X_g. We know that $H_0(X_g)$ is equal to the direct sum of $\mathbb{Z}$ indexed by the path-components of X. Since X_g has one path-component, we have $H_0(X_g) = \mathbb{Z}$.

To compute H_1, we show that the connected sum of g copies of $\mathbb{RP}^2$ can be equipped with the structure of a $2g$-sided polygon with sides identified cyclically, as in the following picture for $\mathbb{RP}^2 \# \mathbb{RP}^2 \# \mathbb{RP}^2$.



If v is the vertex in this connected sum, we may extract a copy of D^2 with its vertex at v, then observe that this attaches via the following connecting map to a copy of $\mathbb{RP}^2$. Since the base case is already satisfied by the definition of $\mathbb{RP}^2$ as its standard CW structure, we have our desired result.

$$\begin{tikzcd}
 & \mathbb{Z}f \arrow[d, leftrightarrow] & \bigoplus_{i=1}^{g} \mathbb{Z}a_i \arrow[d, leftrightarrow] & \mathbb{Z}v \arrow[d, leftrightarrow] & \\
0 \arrow[r, "\partial_3"] & C_2(X_g) \arrow[r, "\partial_2"] & C_1(X_g) \arrow[r, "\partial_1"] & C_0(X_g) \arrow[r, "\partial_0"] & 0 \\
 & & a_i \arrow[r, mapsto] & 0 & \\
 & f \arrow[r, mapsto] & 2\mathbb{Z}(a_1 + \cdots + a_g) & &
\end{tikzcd}$$

In particular, this means that $C_1(X_g) = \bigoplus_{i=1}^{g} \mathbb{Z}a_i$. We observe that for each a_i, $\partial_1(a_i) = v - v = 0$, so that $\ker(\partial_1) = C_1(X_g)$.

Meanwhile, we observe that our face f maps under the boundary map to $2\mathbb{Z}(a_1 + \cdots + a_g)$. In particular, from some linear algebra, we find that

$$\begin{aligned} H_1(X_g) &= \left(\bigoplus_{i=1}^{g} \mathbb{Z}a_i\right) / (2\mathbb{Z}(a_1 + \cdots + a_g)) \\ &\cong \mathbb{Z}/2\mathbb{Z} \oplus \bigoplus_{i=1}^{g-1} \mathbb{Z}. \end{aligned}$$

As for H_2, since $\ker(\partial_2) = 0$ and $\operatorname{im}(\partial_3) = 0$, it follows that $H_2(X_g) = 0$.

Problem (Problem 2)**:** Combined with examples from class, we have computed the homology groups of all closed surfaces. What do you observe about the homology groups of orientable surfaces compared to non-orientable ones?

Solution: Orientable surfaces appear to have a nontrivial 2-dimensional homology group, while non-orientable surfaces don't, while non-orientable surfaces have a copy of $\mathbb{Z}/2\mathbb{Z}$ in the 1-dimensional homology group and have no nontrivial 2-dimensional homology group.